

Jake van Halder

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Education

Washington State University

Bachelor of Science in Computer Science | Graduated May 2026

Experience

Software Engineer Intern, Schweitzer Engineering Laboratories – Pullman, WA

April 2025 – Present

- Developed an HTML/CSS/JavaScript password policy reporting dashboard using Node.js, Shell, and Jenkins, reducing manual review time for 200+ devices across 7 vendors.
- Prevented future UI regressions by adding E2E tests for critical workflows using Playwright and Vitest.
- Improved stability of a customer-facing workflow application by delivering React (TypeScript) features and fixes.
- Collaborated in Agile sprint planning, daily standups, and peer code reviews while supporting CI/CD workflows within a multi-developer codebase.

Frontend Developer, Washington State University – Pullman, WA

October 2024 – May 2026

- Led frontend development for a containerized React (TypeScript) GIS platform serving 10+ organizations, providing interactive access to spatial and tabular data across 126 watersheds.
- Worked closely with researchers and forest-management stakeholders to gather requirements, design Figma wireframes, and coordinate feature delivery in weekly standups.
- Achieved over 85% branch coverage by implementing 37 Vitest suites across 116 files.

Projects

LCBlitz | Next.js, TypeScript, Biome, Better Auth, NeonDB (Postgres)

- Developed a Next.js frontend for a personalized coding-practice platform, leveraging data from 2,000+ problems, 500k+ contest records, and 20M+ users to recommend relevant challenges.
- Designed relational schemas in NeonDB (Postgres) to handle 500K+ records and drive problem recommendation logic.
- Built end-to-end testing and CI pipelines utilizing Playwright, Vitest, and GitHub Actions to validate core user workflows and prevent regressions.

Zuno | React Native Expo, FastAPI, PostgreSQL, Gemini

- Built a React Native Expo frontend for an AI-powered study aid app leveraging Gemini to generate personalized study materials from multimodal user input.
- Designed a reusable component library across 15 screens, enabling faster feature development.
- Worked directly with 5 startup stakeholders to refine requirements, develop Figma wireframes aligned with branding guidelines, and iterate on features based on client feedback.

CGDJ-BulletHell | C#, MonoGame, Git/GitHub

- Built reusable game framework components, including base entity classes and shared infrastructure, improving code reuse and simplifying feature development.
- Architected 11 modular core game systems, enabling parallel development and reducing integration bottlenecks.
- Coordinated development across project milestones leading a 4-person team to deliver a 2D shooter in 4 months.

Skills

Languages: TypeScript, JavaScript, HTML, CSS

Frameworks & Tools: React, Next.js, Git, Docker, Vitest, Playwright, Figma, Jenkins